

RAFFLES INSTITUTION
2017 Year 6 Preliminary Examination

Higher 2

CANDIDATE
NAME

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CIVICS
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BIOLOGY

9744/03

Paper 3 Long Structured and Free-response Questions

14th September 2017

2 hours

Candidates answer on the Question Paper.
Additional Materials: Writing paper.

READ THESE INSTRUCTIONS FIRST

Write your index number, CT group & name in the spaces at the top of this page.

Write in dark blue or black pen.

You may use a HB pencil for any diagrams or graphs.

Do not use staples, paper clips, glue or correction fluid.

DO **NOT** WRITE IN ANY BARCODES

Section A

Answer **all** questions in the spaces provided on the Question Paper.

Section B

Answer any **one** question in the writing paper provided.

The use of an approved scientific calculator is expected, where appropriate.

You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, **hand in your essay question SEPARATELY.**

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
Section A	
1	/ 23
2	/ 27
Section B	
3 or 4	/ 25
Total	/ 75

This document consists of **13** printed pages.



Raffles Institution
Internal Examination

1 (a) Epidermal growth factor (EGF) stimulates cell division and differentiation. Fig. 1.1 shows a cell signaling pathway triggered by EGF.

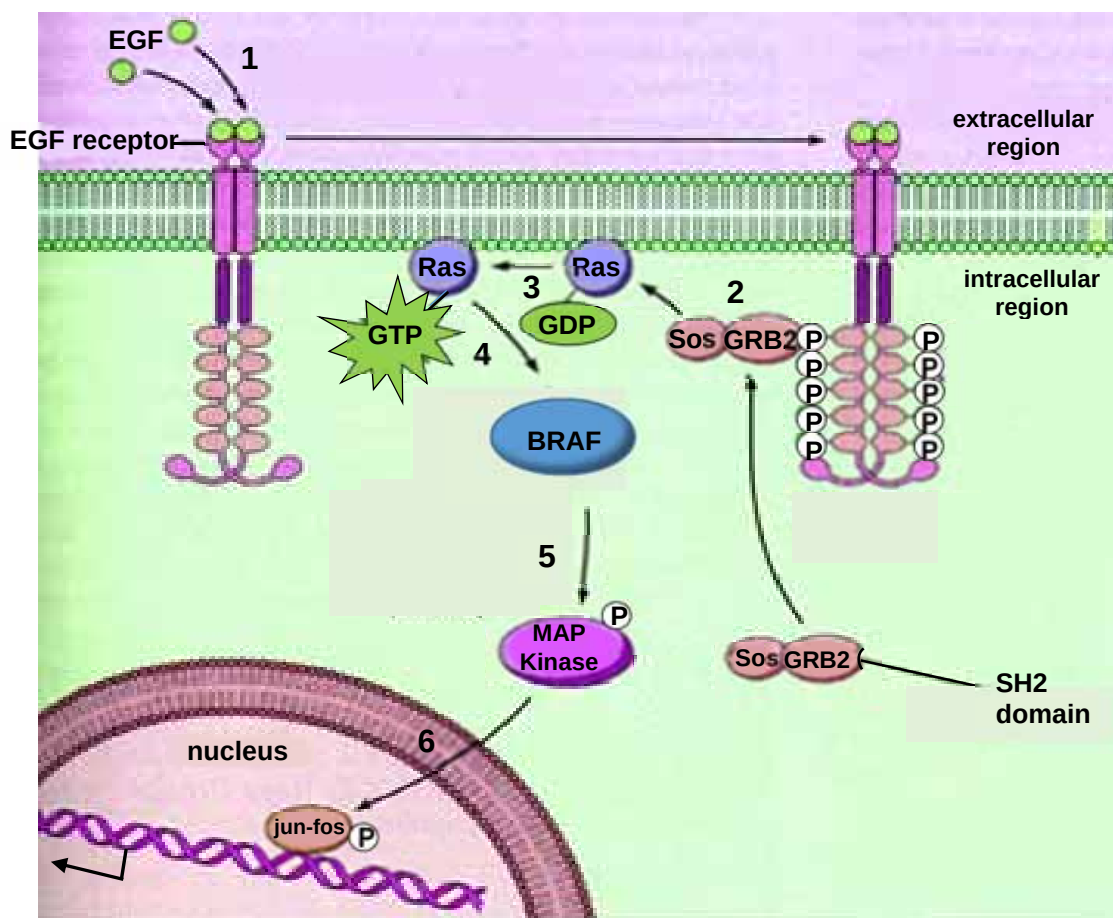


Fig. 1.1

- (i) With reference to Fig.1.1, describe stages 1 to 5.

.....[5]

- (ii) With reference to Fig. 1.1, describe the cellular response caused by EGF.

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.....[2]

- (iii) Cell signaling pathways are advantageous to multicellular organisms.

Describe two advantages of such a signalling pathway.

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.....[2]

- (b) Lung cancer remains as one of the most aggressive cancer types with nearly 1.6 million new cases reported worldwide each year. Chronic exposure to carcinogens results in gene damage in lung epithelial cells causing them to become tumour cells.

Presence of EGF leads to apoptosis inhibition and cell proliferation. Mutations in at least 1 of the genes coding for the proteins shown in Fig. 1.2 have been found in tumour cells of lung cancer patients.

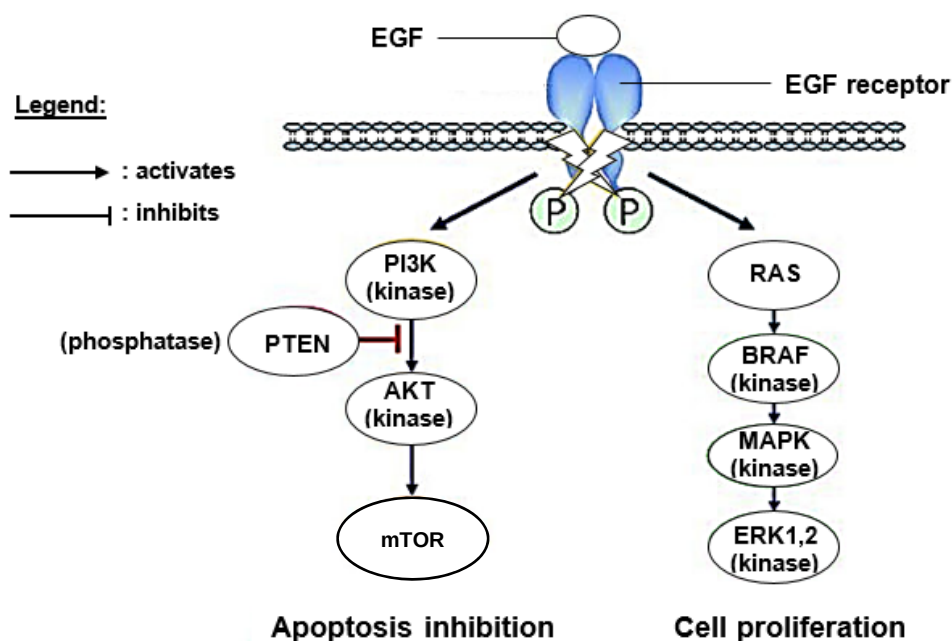


Fig. 1.2

- (i) Name a carcinogen that can increase the chances of cancerous growth in the lungs.

.....[1]

- (ii) Assuming the nucleotide sequence of the *Ras* gene was unchanged, suggest and explain one way in which a gain-in-function mutation of the *Ras* gene can occur.

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.....[2]

- (iii) With reference to Fig. 1.2, suggest how *PTEN* functions as a tumour suppressor gene.

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.....[2]

- (c) In stem and cancer cells, telomerase is activated to prevent the shortening of telomeres to a critical length.

The Hayflick Limit is the number of times a somatic cell will divide until cell division stops. Telomeres associated with each cell's DNA will shorten with each cell division until they shorten to a critical length.

- (i) Describe the role of telomerase in cancer and stem cells.

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.....[3]

- (ii) Telomerase activity is reactivated in tumour cells.

Suggest and explain how telomerase can be inhibited as a form of cancer therapy.

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.....[2]

- (d) Fig. 1.3 shows the development of an embryonic stem cell into a somatic cell, and a somatic cell being reprogrammed into an induced pluripotent stem cell (iPSC).

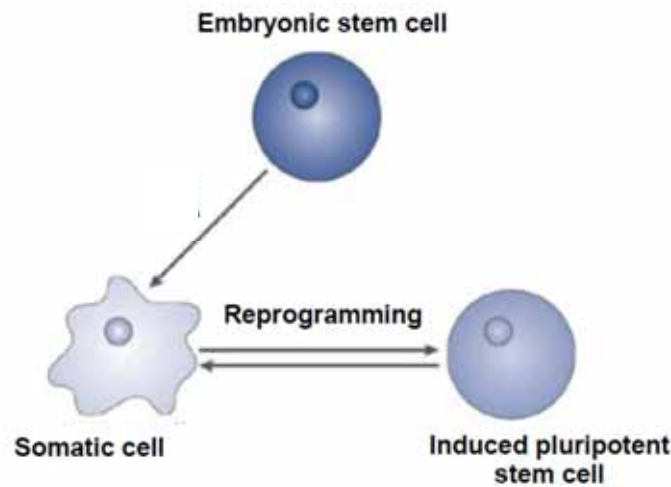


Fig. 1.3

Embryonic stem cells have been known to express a specific set of genes that enable them to behave like stem cells, but these genes are turned off in normal somatic cells. Researchers have discovered that they can convert human fibroblast cells into embryonic stem cells by switching on the specific set of genes. These cells have been cultured in labs all over the world, and they are termed induced pluripotent stem cells (iPSCs).

- (i) Describe the characteristics of an induced pluripotent stem cell (iPSCs).

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.....[2]

- (ii) Traditionally, the only source of embryonic stem cells is from the inner cell mass of an embryo, thus raising many ethical issues with the use of stem cells from embryos. The discovery of iPSC is exciting for scientists, as it enables research to be carried out on cells that behave like embryonic stem cells, without obtaining them from embryos.

Suggest two ethical concerns regarding the use of embryonic stem cells which are no longer relevant with the development of iPSCs.

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.....[2]

[Total: 23]

- 2 Dengue fever is a disease spread by a particular species of mosquito, *Aedes aegypti*. The incidence of dengue has dramatically increased in recent years. This has heightened the need to understand the vector, as well as the virus.

- (a) Fig. 2.1 shows the distribution of dengue fever in 1990, as well as the estimated distribution of dengue fever in 2085.

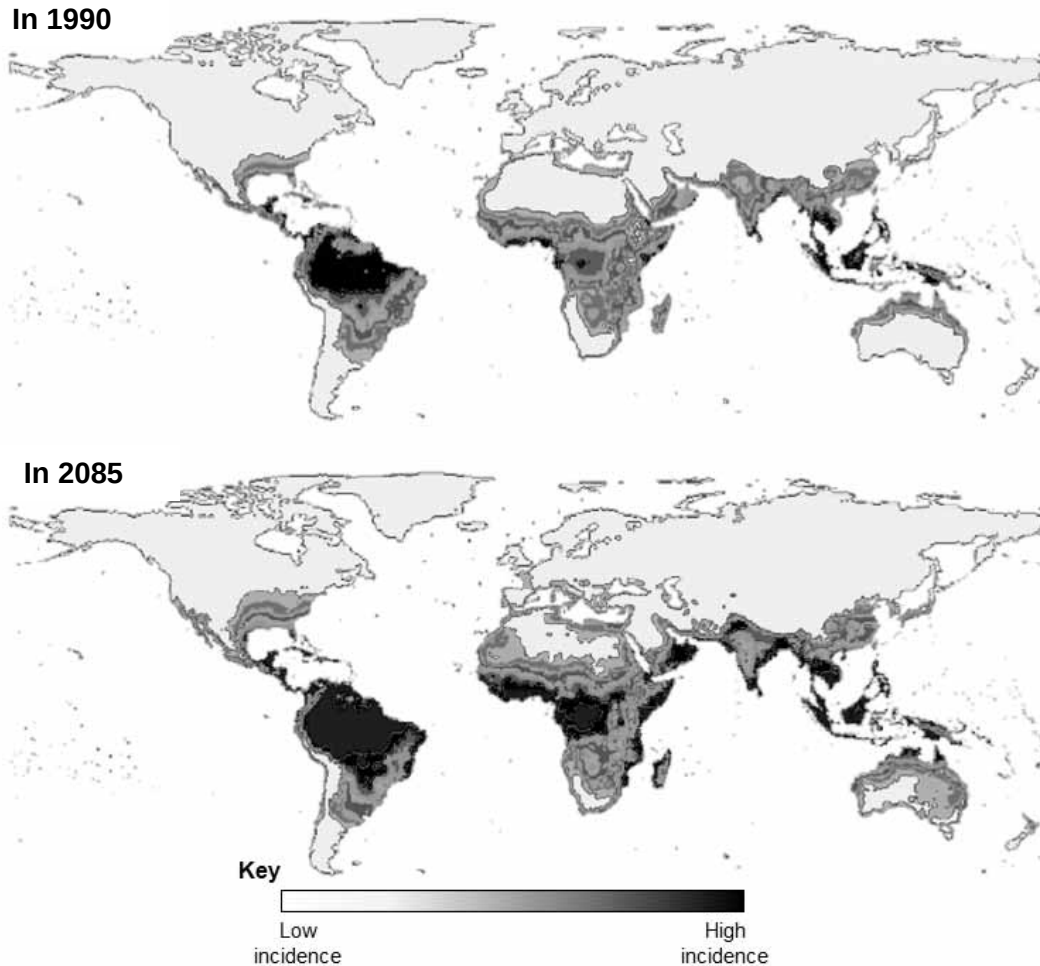


Fig. 2.1

- (i) With reference to Fig. 2.1, describe and explain how climate change may influence the distribution of dengue fever.

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..... [3]

In order to arrive at the 2085 prediction in Fig. 2.1, scientists have looked into the factors affecting the mosquito's physiology and fitness in transmitting the disease. Fig. 2.2 shows the summary of their findings.

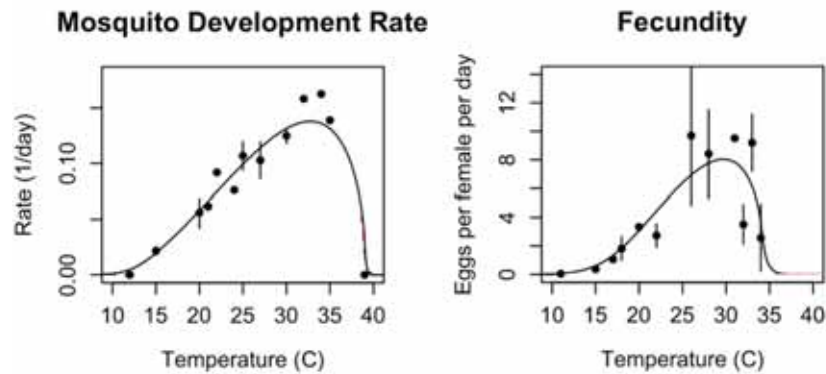


Fig. 2.2

With reference to Fig. 2.2,

- (ii) describe how the change in temperature from 20°C to 30°C may affect the transmission of dengue.

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 [2]

- (iii) predict and explain the transmission of dengue at 40°C.

.....

 [2]

(b) The dengue virus spends part of its life cycle living in mosquitoes and another part of its life cycle living in humans where it causes the disease, dengue fever.

(i) State the name of the type of interaction between dengue virus and its human host.

..... [1]

(ii) Explain the difference between an infectious and a non-infectious disease.

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..... [2]

In order to design a treatment for dengue fever, the reproductive cycle of the dengue virus was and is being studied. The dengue virus is found to be in the family of *Flavivirus*, which has characteristics similar to both the influenza virus and HIV.

Fig. 2.3 shows the structure of a dengue virus which contains a positive stranded RNA.

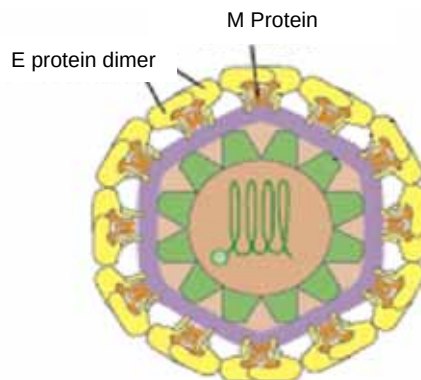


Fig. 2.3

(iii) State one similarity that the dengue virus shares with both influenza virus and HIV.

.....[1]

- (c) Fig. 2.4 shows the reproductive cycle of the dengue virus in a human host cell after an individual was bitten by an *Aedes* mosquito carrying the virus.

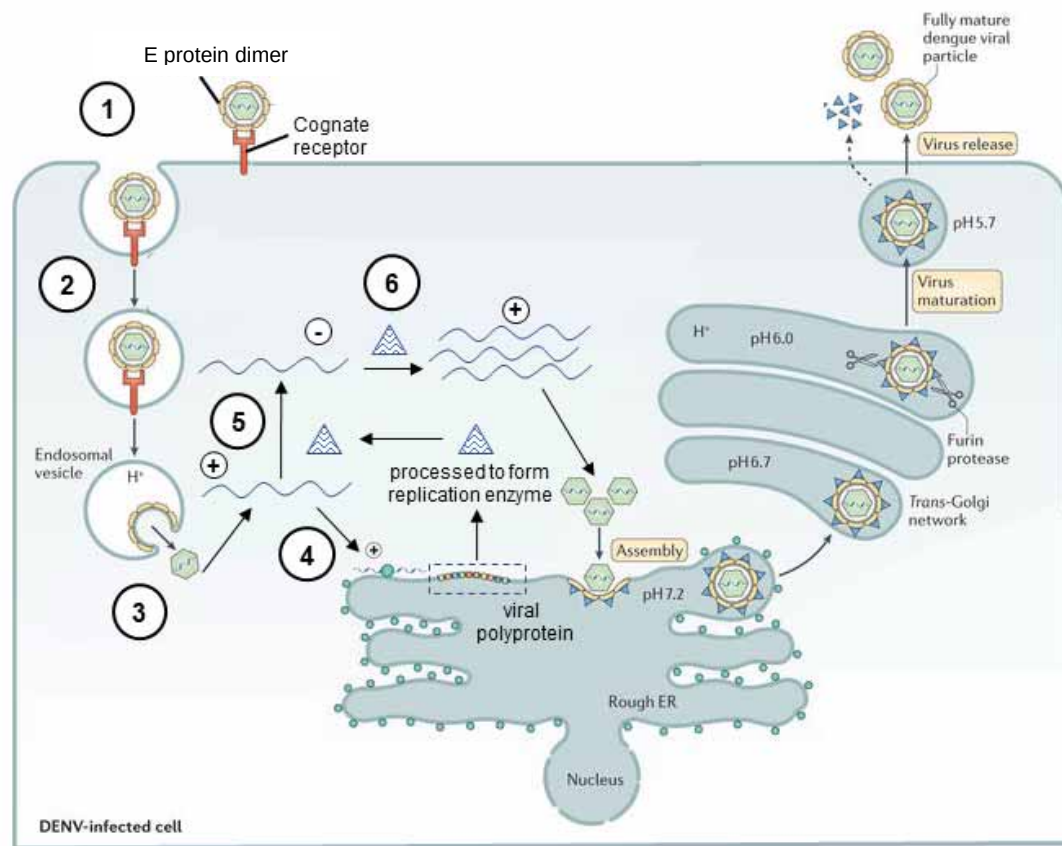


Fig. 2.4

Adapted from Nature Immunology

With reference to Fig. 2.4, describe the following stages in the life cycle of the dengue virus.

- (i) stages 1 to 3

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..... [3]

- (ii) stages 4 to 6

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- (iii) With reference to Fig 2.4, suggest how researchers may design a drug to prevent the replication of dengue virus within a human host cell.

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..... [1]

- (d) A patient is required to go for a blood test to find out if he has been infected by the dengue virus. The blood test will determine the viral load in a person's blood stream.

Currently, there are no known medical treatments for dengue. Patients are recommended to rest and let the body's immune system eliminate the virus.

- (i) Describe how the patient's specific immune system is able to remove cells infected by the dengue virus.

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..... [2]

- (ii) Dengvaxia® is a vaccine that is currently being administered as a measure of dengue prevention in various countries stricken by dengue epidemics.

Explain how the vaccine confers an individual protection against the dengue virus.

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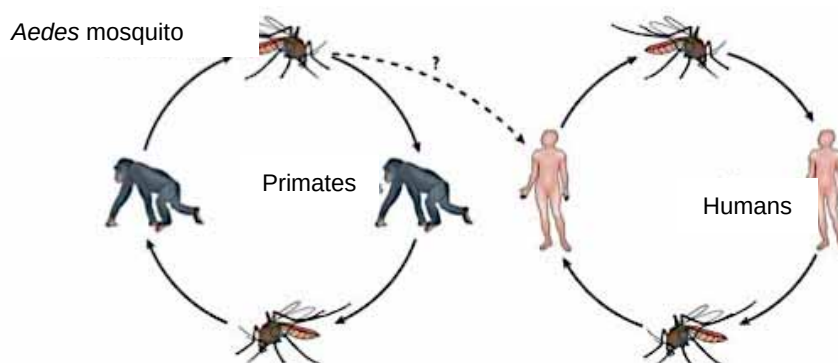
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..... [5]

- (e) The abstract in Fig. 2.5 was taken from an article in Nature Immunology.



"Scientists hypothesize that the dengue viruses evolved in nonhuman primates and jumped from these primates to humans in Africa or Southeast Asia between 500 and 1,000 years ago."

Fig. 2.5

Using your knowledge of viruses, explain the changes in the dengue virus that could have led to the phenomenon in the statement above.

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..... [2]

[Total: 27]

Section B

Answer **one** question in this section.

Write your answers on the separate answer paper provided.

Your answers should be illustrated by large, clearly labeled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

- 3** **(a)** Describe how variation arises and how recessive alleles are preserved in a population. [13]
 (b) Compare and contrast the techniques of polymerase chain reaction and Southern hybridisation. [12]

[Total: 25]

- 4** **(a)** Describe how anatomical and molecular homology support Darwin's theory of descent with modification. [13]
 (b) Describe the effects of different types of mutations on the proteins of eukaryotes. [12]

[Total: 25]

End of Paper